

Conformer + CTC Architecture — 9.4M parameters

(A) Input pipeline

Input video → 86-joint skeleton pose
(T frames, 86 joints, 2 coords)



Body-centric normalization + std scaling



Flatten + velocity features → 344-dim input



Augmentation: time warp, rotation,
scale, joint dropout, frame mask



(B) Conformer encoder

Linear projection → d_model=256 + sinusoidal PE



6 × Conformer Block
(Macaron FFN → MHSA → Conv → FFN → LN)
d_model=256, heads=4, conv_kernel=31



Linear (256 → 678) → log-softmax



(C) CTC head

CTC decode → Arabic gloss sequence